02443 Stochastic Simulation

Day 5: Markov Chain Monte Carlo (MCMC)

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Agenda

1. Introduction
2. Example from Bayesian statistics
3. The Metropolis-Hastings algorithm
4. Gibbs sampling
5. Exercises

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The problem

Suppose we have a random variable X with density $f(x) = cg(x)$, where

- the function $g(x)$ can be evaluated,
- the normalizing constant c is unknown or difficult to compute.

As a result, we cannot easily generate independent samples from the distribution of X.

Typical applications

- Bayesian posterior distributions
- High-dimensional probability distributions
- Machine learning models
- Statistical physics

Our goal is to generate samples from the target distribution despite not knowing the normalizing constant.

The basic idea of MCMC

Recall the previous lecture:

Given a Markov chain, we determined its stationary distribution.

In Markov Chain Monte Carlo we reverse this logic:

Given a target distribution, we construct a Markov chain whose stationary distribution is exactly the target distribution.

We then simulate the Markov chain and use the generated states as samples from the target distribution.

The price we pay is that the generated samples are no longer independent.

Overview of MCMC methods

Methods covered in this lecture

- Metropolis-Hastings algorithm
- Random Walk Metropolis-Hastings
- Coordinatewise Metropolis-Hastings
- Gibbs sampling

General strategy

1. Construct a Markov chain with the desired stationary distribution.
2. Simulate the Markov chain.
3. Discard an initial burn-in period.
4. Use the remaining states as samples from the target distribution.

Throughout the lecture we will focus on how to construct such Markov chains and how to verify that they have the desired stationary distribution.

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Example: Bayesian inference

Suppose we wish to estimate an unknown probability parameter P. We are given that conditional on $P = p$, the random variable X follows a binomial distribution with parameters n and p:

$$P(X = i \mid P = p) = \binom{n}{i} p^i (1-p)^{n-i}.$$

Prior distribution

Before observing any data, we assume $P \sim U(0,1)$ (an uninformed prior distribution) with density

$$f_P(p) = 1, \qquad 0 \le p \le 1.$$

Observed data

Suppose we observe i successes in n independent trials.

Our objective is to determine the distribution of the parameter P after observing the data.

The posterior distribution

Using Bayes' theorem, the posterior density of P is

$$f_{P|X}(p|X=i) = \frac{f_P(p)\,P(X=i \mid P=p)}{P(X=i)}.$$

The denominator is given by

$$P(X=i) = \int_0^1 f_P(p)\binom{n}{i}p^i(1-p)^{n-i}\,dp.$$

Key observation

The posterior density can be written as

$$f_{P|X}(p|X=i) = c f_P(p)\,P(X=i \mid P=p),$$

where

$$c = \frac{1}{P(X=i)}$$

is a normalizing constant.

In this simple example, the integral can be evaluated analytically. In more complex models, this is often impossible, and MCMC provides a way to sample from the posterior distribution without evaluating the normalizing constant.

The genius idea

How can we sample from a distribution we do not know?
Initially, this sounds like an impossible task.

The trick:
We do not know $f(x)$, but we do know $f(x)/f(y) = g(x)/g(y)$ for all x and y.

In terms of the Markov chain:
We do not construct a Markov chain from the equations governing the stationary distribution directly, but rather construct a Markov chain satisfying the local balance equations.

We construct a transition kernel such that

$$f(i)p_{ij} = f(j)p_{ji} \Longrightarrow cg(i)p_{ij} = cg(j)p_{ji} \Longleftrightarrow g(i)p_{ij} = g(j)p_{ji}.$$

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The Metropolis-Hastings algorithm

Suppose we wish to sample from a target density

$$f(x) = c\,g(x),$$

where the normalizing constant c is unknown.

Basic idea

Starting from the current state x:

1. Propose a candidate state y.
2. Decide whether to accept or reject the proposal.
3. If accepted, move to y.
4. Otherwise remain at x.

Repeating this procedure generates a Markov chain

$$X_0, X_1, X_2, \ldots$$

whose stationary distribution will be the target distribution.

Proposal distributions

The proposal is generated from a proposal distribution $h(x, y)$, where

$$h(x, y) = \mathbb{P}(X_{n+1} = y \mid X_n = x).$$

Interpretation

- x is the current state.
- y is the proposed state.
- $h(x, y)$ determines how we explore the state space.

A good proposal distribution should:

- Be easy to sample from.
- Be easy to evaluate.
- Explore the state space efficiently.
- Avoid excessive rejection of proposals.

Acceptance probability

Once a candidate state y has been proposed, it is accepted with probability

$$\alpha(x,y) = \min\left(1, \frac{f(y)h(y,x)}{f(x)h(x,y)}\right).$$

If the proposal is accepted:

$$X_{n+1} = y.$$

Otherwise:

$$X_{n+1} = x.$$

Important observation

Since

$$f(x) = c\,g(x),$$

the acceptance probability becomes

$$\alpha(x,y) = \min\left(1, \frac{g(y)h(y,x)}{g(x)h(x,y)}\right).$$

The normalizing constant cancels completely.

Symmetric proposal distributions

A common choice is a symmetric (in x and y) proposal distribution satisfying

$$h(x,y) = h(y,x).$$

In this case the acceptance probability simplifies to

$$\alpha(x,y) = \min\left(1, \frac{g(y)}{g(x)}\right).$$

Interpretation

- Moves toward regions of higher density are always accepted.
- Moves toward regions of lower density are accepted with a probability less than one.

This special case is known as the **Metropolis algorithm**.

Why does Metropolis-Hastings work?

The transition probability from x to y is

$$p(x,y) = h(x,y)\min\left(1, \frac{g(y)h(y,x)}{g(x)h(x,y)}\right).$$

Assume first that

$$g(y)h(y,x) < g(x)h(x,y).$$

Then

$$p(x,y) = h(x,y)\frac{g(y)h(y,x)}{g(x)h(x,y)}.$$

Why does Metropolis-Hastings work? (continued)

Multiplying by the target density gives

$$f(x)p(x,y) = cg(x)h(x,y)\frac{g(y)h(y,x)}{g(x)h(x,y)} = cg(y)h(y,x).$$

Since

$$p(y,x) = h(y,x),$$

we obtain

$$f(x)p(x,y) = cg(y)h(y,x) = f(y)p(y,x).$$

Thus the local balance equation is satisfied.

The opposite case is completely analogous.

Why the Metropolis-Hastings works (summary)

The Metropolis-Hastings acceptance probability is chosen so that

$$f(x)p(x,y) = f(y)p(y,x).$$

Consequently, the target distribution satisfies the local balance equations and is therefore a stationary distribution of the Markov chain.

Key observation

The proof only involves ratios of the target density:

$$\frac{f(y)}{f(x)} = \frac{g(y)}{g(x)}.$$

The unknown normalizing constant therefore cancels automatically.

This allows us to sample from distributions that are only known up to proportionality.

Random Walk Metropolis-Hastings

A simple and widely used proposal mechanism is the **random walk proposal**.

Suppose the current state of the Markov chain is X_i.

Generate a random perturbation Δ_i from a symmetric distribution. Then propose the state:

$$Y_i = X_i + \Delta_i.$$

Since the distribution of Δ_i is symmetric, we have

$$h(x,y) = h(y,x).$$

Consequently, the Metropolis-Hastings acceptance probability simplifies to

$$\alpha(x,y) = \min\left(1, \frac{g(y)}{g(x)}\right).$$

Random Walk Metropolis-Hastings Algorithm

Suppose the current state is X_i.

1. Generate a proposal $Y_i = X_i + \Delta_i$.
2. If $g(Y_i) \geq g(X_i)$, accept the proposal.
3. Otherwise, accept the proposal with probability $\frac{g(Y_i)}{g(X_i)}$.
4. If the proposal is accepted, set $X_{i+1} = Y_i$.
5. Otherwise, set $X_{i+1} = X_i$.

The resulting Markov chain has the desired target distribution as its stationary distribution.

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Gibbs sampling

Suppose we wish to sample from a multivariate distribution

$$f(x_1, \ldots, x_d).$$

Assume that the conditional distributions are known and easy to sample from:

$$f(x_1 \mid x_2, \ldots, x_d),$$

$$f(x_2 \mid x_1, x_3, \ldots, x_d),$$

$$\vdots$$

$$f(x_d \mid x_1, \ldots, x_{d-1}).$$

The basic idea is to update one coordinate at a time while keeping the remaining coordinates fixed.

The resulting Markov chain has the desired joint distribution as its stationary distribution.

Gibbs sampling algorithm

Suppose the current state is

$$\boldsymbol{X}^{(i)} = (x_1^{(i)}, \dots, x_d^{(i)}).$$

One Gibbs iteration consists of:

1. Sample
$$x_1^{(i+1)} \text{ from } f(x_1 \mid x_2^{(i)}, \dots, x_d^{(i)}).$$
2. Sample
$$x_2^{(i+1)} \text{ from } f(x_2 \mid x_1^{(i+1)}, x_3^{(i)}, \dots, x_d^{(i)}).$$
3. Continue until all coordinates have been updated.

Important difference from Metropolis-Hastings

No acceptance-rejection step is required.
Every proposed value is accepted because it is generated directly from the correct conditional distribution.

MCMC in perspective

Recall the previous lecture on Markov chains.

Ordinary Markov chains

We are given a transition kernel $P(x,y)$ and seek to determine the stationary distribution π. This can be done analytically or through simulation.

Markov Chain Monte Carlo

We are given a target distribution$f(x) = g(x)$ and seek to construct a transition kernel $P(x,y)$ having f as its stationary distribution.

Simulation of the resulting Markov chain then provides samples from the target distribution.

Practical considerations

Markov Chain Monte Carlo methods are extremely powerful, but they also present a number of challenges.

Important considerations

- The generated samples are dependent.
- The chain may require a substantial burn-in period.
- Convergence is difficult to assess in practice.
- Different starting values may lead to different transient behaviour.
- The efficiency depends strongly on the choice of proposal distribution.

Common practice

- Run multiple chains from different starting points.
- Compare within-chain and between-chain variability.
- Inspect trace plots and empirical distributions.
- Discard an initial burn-in period.

Beyond Metropolis-Hastings

The methods presented in this lecture are only the beginning.

Possible extensions

- More sophisticated proposal distributions.
- Adaptive Metropolis-Hastings algorithms.
- Hybrid and problem-specific MCMC algorithms.

Many modern Bayesian and machine learning applications rely heavily on advanced MCMC techniques.

Key takeaway

The central idea remains unchanged: Construct a Markov chain whose stationary distribution is the desired target distribution.

Everything else concerns how efficiently we can explore that distribution.

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Exercise 6

Part 1

The number of busy lines in a trunk group (Erlang system) follows a truncated Poisson distribution

$$P(i) = c\frac{A^i}{i!}, \qquad i = 0, \ldots, m.$$

Generate samples from this distribution using the Metropolis-Hastings algorithm.

- Verify the distribution using a χ^2 goodness-of-fit test.
- You may use the parameter values from Exercise 4.

Part 2

For two different call types, the joint number of occupied lines is given by

$$P(i,j) = c\frac{A_1^i}{i!}\frac{A_2^j}{j!}, \qquad 0 \leq i+j \leq m.$$

Use

$$A_1 = A_2 = 4, \qquad m = 10.$$

Exercise 6 (continued)

For the distribution

$$P(i,j) = c\frac{A_1^i}{i!}\frac{A_2^j}{j!}, \qquad 0 \leq i+j \leq m,$$

complete the following tasks:

(a) Use the Metropolis-Hastings algorithm directly to generate samples from the distribution.

(b) Use coordinatewise Metropolis-Hastings to generate samples from the distribution.

(c) Use Gibbs sampling to generate samples from the distribution.

Determine the required conditional distributions analytically and use them in the Gibbs sampler.

In all three cases, assess the fit using a χ^2 goodness-of-fit test.

Remark: The model can easily be extended to higher dimensions and additional restrictions on the different call types can be incorporated.

Exercise 6 (continued)

Part 3

Consider the Bayesian model

$$X_i \sim N(\Theta, \Psi),$$

where the prior distribution of

$$(\Xi, \Gamma) = (\log(\Theta), \log(\Psi))$$

is bivariate normal with standard normal marginals and correlation $\rho = \frac{1}{2}$. The joint density of (Θ, Ψ) is

$$f(x,y) = \frac{1}{2\pi xy\sqrt{1-\rho^2}} \exp\left(-\frac{\log(x)^2 - 2\rho\log(x)\log(y) + \log(y)^2}{2(1-\rho^2)}\right).$$

(a) Generate a sample (θ, ψ) from the prior distribution by first generating a sample (ξ, γ) from (Ξ, Γ).

(b) Using the generated values (θ, ψ), simulate observations $X_1, \ldots, X_n$ with $n = 10$.

Exercise 6 (continued)

(Continue Part 3).

(c) Derive the posterior distribution of (Θ, Ψ) given the observed sample.
Hint: Apply Bayes' theorem in its density form.

(d) Generate samples from the posterior distribution using the Metropolis-Hastings algorithm.

(e) Repeat Part (d) with $n = 100$ and $n = 1000$. Use the same values of (θ, ψ) generated in Part (a) and discuss the results. "'

Remark

The sample mean and sample variance are independent. The sample mean follows a normal distribution, while a suitably scaled sample variance follows a χ^2 distribution. This observation can be used to simplify the posterior distribution and reduce the computational effort.